

CHARCUBE · REQUIREMENTS

CharCube requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/charcube		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (CCB-DDR-001); R1 limited to the decided feedstock, R10 redefined to exclude the required safety kit, R9 probe class stated; status from CCB-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

CharCube requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. R1 and R10 were redefined by Amish's decisions of 2026-09-25 (CCB-DDR-001 items 2 and 4). The status column gives the TRL 3 result from CCB-CAL-001 (central estimate, with the range where it matters): met, not met, at risk, or not verifiable at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CCB-CAL-001)
R1	Process a useful batch	10 kg or more of air-dry residue (15 % moisture or less) per batch, for the decided feedstock: bundled straw and mixed stalks and cobs, packed to about 120 kg/m³ . Loose straw is out of scope without a press (a separate project)	Retort volume and packed bulk density	Met: 12.0 kg (10.58 kg dry)
R2	Make biochar efficiently	Biochar 25 % or more of dry feed mass	Mass balance; later weighed batches	At risk: 28 % (20 to 35 %), 2.96 kg
R3	Make stable char	Retort core at 450 °C or more for 30 min or more per batch, logged; target molar H/Corg below 0.7, the European Biochar Certificate limit	Heat conduction and balance; later logged batches and lab analysis	At risk: core at 450 °C about 4.0 h after lighting (2.9 to 6.5 h), only if the annulus is kept hot
R4	Burn the gas, not vent it	All retort gas passes the annulus fire and the burner throat; visible smoke only during the first 15 min after lighting and the last 10 min of a batch; methane below 24 g per kg of char (the retort field average in Sparrevik et al. 2015)	Design review and air sizing; later emission measurement with a partner lab	Not verifiable at TRL 3: route closed, draft margin 3.2, throat holes sized with 15 % spare area
R5	Fit a daily routine	Light to end of flaming in 4 h or less; char ready to unload within 16 h of lighting	Heat conduction and cooling; later logged batches	Not met: burn about 4.5 h (3.4 to 7.0 h); unload after about 8 h is met
R6	Recover heat as hot water	10 MJ or more into water per batch (60 L raised by 40 K)	Heat transfer; later measured water temperature	Not met: 6.4 MJ (4.9 to 9.2 MJ), about 25 K rise temperature
R7	Keep the water circuit safe	Jacket open to the air at all times; no valve or cap that can seal it; water drawn from a tap at the base; jacket kept away from the lid by a separate support	Design review and safety checklist	Met by design
R8	Use little start-up fuel	5 kg or less of dry wood per batch	Energy balance; later weighed batches	Not met: 5.8 kg (2.9 to 13.9 kg); light-up alone 1.7 kg
R9	Record each batch	Two type K channels (retort core, throat exit), class 1 probes, 0 to 1,000 °C, ±5 °C or better, logged every 10 s to removable storage, 24 h on a USB power bank	Component data; design review	At risk: ±3.4 °C to 700 °C; amplifier accuracy not specified above 700 °C; energy 6 of 31 Wh
R10	Stay within the concept budget	Kiln parts \$300 or less. The safety kit (extinguisher or water and shovel, gloves, eye protection, dust mask, about \$40) is required and listed separately, outside the kiln budget	Priced BOM	Met: \$298
R11			Masses from the model; design review	Met: lift unit 43.3 kg for two people (21.6 kg

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CCB-CAL-001)
	Be built and handled by a small crew	Built with hand tools, drill and angle grinder, with at most one welded part (the water jacket); no single lift above 25 kg per person		each); retort with char 15.2 kg
R12	Operate safely outdoors	Flue outlet 2.5 m or more above the ground; 5 m clear radius from structures and dry vegetation; no galvanized parts in the hot path	Model and BOM; safety checklist	Met: outlet 2.70 m; no galvanized lines

■ Requirements not met or at risk

- **R5 not met.** Heat conducts slowly into a 0.46 m diameter packed charge; the burn lasts about 4.5 h. Proposed, awaiting Amish: relax R5 to 5 h (CCB-DDR-001 item 14).
- **R6 not met.** A plain flue sleeve transfers about 4 to 5 W/m²K, so the jacket takes about 6.4 MJ, not the 15 MJ estimated at TRL 2. A baffle insert and a jacket blanket would give about 13.9 MJ on paper. Proposed, awaiting Amish (item 12).
- **R8 not met.** Keeping the annulus hot for the whole burn needs about 5.8 kg of wood. Proposed, awaiting Amish: blanket the lid and keep the target until batches are logged (item 13).
- **R2, R3 and R9 at risk.** Yield and core temperature depend on feedstock and operation; R9 needs class 1 probes and has no amplifier specification above 700 °C (item 15).
- **R4** cannot be shown without emission measurement.
- **R11** is met only with the jacket drained and two people lifting the tripod unit.

■ Assumptions

- Feedstock (decided): bundled straw, maize or cotton stalks, cobs or prunings at 10 to 15 % moisture (12 % in CCB-CAL-001), packed to about 120 kg/m³; about 18.2 MJ/kg dry and 16.0 MJ/kg air-dry (higher heating value).
- Start-up fuel: air-dry wood at about 16.2 MJ/kg (lower heating value), preferably from residues or prunings.
- Char: about 20 MJ/kg and about 55 % organic carbon with about 29 % ash for mixed straw and stalks (IPCC 2019 defaults are 49 % for rice straw char and 65 % for herbaceous material).
- One batch a day for about 250 days a year.
- Ambient 10 to 40 °C, dry weather, wind below about 5 m/s.
- The kiln serves a household's or a trial plot's residue, not a whole field (see CCB-PRB-001, out of scope).